Dmytro Barannik, PhD

Staff Software Engineer & Tech Lead

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EXECUTIVE PROFILE

Staff Software Engineer, Tech Lead & Systems Architect with **9+ years of engineering leadership** spanning high-load streaming media, autonomous AI swarm orchestration ("vibe-coder & architect"), zero-allocation C++20 systems, multiplatform mobile & edge web, and applied cryptographic compression research. Proven track record scaling client architectures for millions of active users at **MEGOGO**, leading enterprise teams across Smart Home & Healthcare at **CHI Software**, bootstrapping indie software ventures to **25,000+ downloads with a 5.0★ rating** (**DiMALiNA**), and inventing state-patented algebraic compression algorithms. Expert at directing multi-agent AI systems, conducting high-level technical reviews, and eliminating architectural debt.

COMMERCIAL EXPERIENCE & TECHNICAL LEADERSHIP

Founder & Principal Software Engineer — DiMALiNA Studio — Commercial Indie — 09.2017 – Present

- Bootstrapped and shipped independent commercial software products with **25,000+ total downloads** and a perfect **5.0★ rating** on Google Play (*Tic-Tac-Toe: 3 Player Game*, *Brainrot Clicker*, *People Notes AI CRM*).
- Owned entire end-to-end engineering lifecycles solo: UX/UI design in Figma, native Jetpack Compose & Clean Architecture, offline-first Room databases, and multi-tier Coroutine flows.
- Architected real-time multiplayer backend pipelines using **Firebase Firestore, Cloud Functions**, and seamless offline data synchronization.
- Engineered commercial monetization models: integrated Google Play Billing (IAP subscriptions/consumables) and optimized AdMob mediation waterfalls.
- Spearheaded multi-agent autonomous AI development workflows ("vibe-coder & architect"), drastically accelerating delivery velocity from conception to store releases.

Senior Software Engineer (Streaming & Android) — MEGOGO — High-Load OTT Media — 03.2025 – 11.2025

- Engineered high-load client streaming pipelines for a premier OTT streaming platform serving **millions of active users** across mobile and TV ecosystems.
- Migrated legacy XML-based screens to modern declarative **Jetpack Compose**, achieving a **~35% boost in UI rendering performance** and eliminating UI jank during rapid catalog scrolling.
- Implemented channel availability, parental locks, and playback entry workflows utilizing **ExoPlayer** with custom low-latency adaptive buffering.
- Maintained **99.8% crash-free sessions** across millions of active streams through rigorous lifecycle scoping, profiler memory leak audits, and automated regression testing.
- Engineered complex deep-linking attribution capabilities via Install Referrer API and customized URI scheme dispatchers, maximizing acquisition funnels.

Lead Software Engineer / Tech Lead — CHI Software — Enterprise Consultancy — 09.2020 – 03.2025

- Spearheaded distributed engineering teams delivering mission-critical applications across Healthcare, Smart Home, Security, and Media domains.
- Integrated high-performance C++ ALPR (Automatic License Plate Recognition) OCR engine into Android via JNI/NDK, optimizing Camera 2 image streams in real time.
- Architected the remote configuration platform **LiftApp** utilizing Wi-Fi Direct and BLE, enabling seamless zero-config peer-to-peer hardware control.
- Orchestrated architectural migrations toward reactive unidirectional data flows (MVI/MVVM) and multi-module structures, reducing incremental build times by **~40%**.
- Mentored 10+ junior and mid developers, authored architectural guidelines, and instituted automated CI/CD quality gates, drastically reducing bug density.

Senior Software Engineer (Mobile & Systems) — Nitrix Studio — Commercial Agency — 08.2018 – 08.2020

- Engineered cross-platform client solutions (Flutter BLoC & Native Android) and backend microservices with real-time WebSocket messaging and SQLite/Room caching.
- Designed scalable domain architectures, authored comprehensive automated testing suites, and standardized deployment automations.

Software Engineer — IT Company — Commercial Software — 09.2017 – 06.2018

- Developed native Android client applications in Java and Kotlin following Material Design guidelines and reactive event patterns.
- Integrated third-party RESTful APIs, location services, and background synchronization workers.

TECHNICAL STACK & ARCHITECTURAL EXPERTISE

AUTONOMOUS AI & ARCHITECTURE
AI Agent Swarms & Orchestration, Prompt Architecture, Clean / Hexagonal / Event-Driven Architectures, System Decomposition, Zero-Regression Verification.

MOBILE & MULTIPLATFORM
Kotlin (Expert), Jetpack Compose, Material 3, Kotlin Multiplatform (KMP), Flutter (Dart), Coroutines, Flow, Room, ExoPlayer, Android NDK/JNI.

SYSTEMS & LOW-LEVEL C++
Modern C++20/23, SIMD (AVX2/NEON), Zero-Allocation Memory Buffers, Memory-Mapped I/O, POSIX/Linux, CMake, Cross-Compilation Toolchains.

WEB, CLOUD & REAL-TIME
Astro Islands, Next.js, Node.js, WebSockets, Firebase Firestore & Cloud Functions, RESTful APIs, SQLite, Sub-Second LCP Web Optimization.

IOT & EMBEDDED TELEMETRY
ESP32, LoRaWAN / LoRa Mesh, BLE (Bluetooth Low Energy), Wi-Fi Direct, MQTT, Microcontroller Firmware, Hardware Sensor Calibration.

INFOSEC & CRYPTOGRAPHY
Algebraic Polynomial Lossless Compression, Steganographic Detection, Cryptographic Protocols, Digital Forensics, Covert Channel Prevention.

ACADEMIC LEADERSHIP & SCIENTIFIC RESEARCH

Associate Professor at Department of Combating Cybercrime KhNUIA Academic & R&D 09.2026 – Present

Lecturing and designing advanced curricula on cybersecurity, digital forensics, and network stream protection. Directing research initiatives in automated incident response and covert channel detection.

State Patent: Polynomial Data Compression | Ukrainian State Patent Office
Sole inventor of a novel algorithmic compression architecture (**LibPolyCodec**), achieving up to 35% lossless compaction with SIMD AVX2 acceleration and zero memory allocation.

Horizon Europe & Scientific Grant Proposals | Applied R&D
Principal investigator on high-impact research proposals in resilient mesh telemetry and secure edge communications. Multiple first-prize awards in national IT research competitions.

ARCHITECTURAL & OPEN-SOURCE HIGHLIGHTS

LiftApp | Lead Architect · Kotlin, Wi-Fi Direct, BLE, Clean Architecture
Engineered peer-to-peer remote configuration software for elevator electronics, achieving zero-configuration hardware discovery in industrial high-interference environments.

LibPolyCodec | Author & Systems Engineer · C++20, SIMD AVX2/NEON
High-performance compression library utilizing algebraic polynomial coefficients, achieving sub-millisecond throughput with zero dynamic allocations in critical inner loops.

LoRa Mesh Network | Embedded Systems Architect · C++, ESP32, LoRa
Designed resilient multi-hop off-grid emergency telemetry mesh network featuring dynamic routing and ultra-low power consumption for field sensor arrays.

ESMEE Bionic Platform | Tech Lead · Astro, WebSockets, C++
Integrated medical and rehabilitation platform providing real-time telemetry, myoelectric sensor calibration, and diagnostic dashboards for upper-limb bionic prostheses.

EDUCATION & CREDENTIALS

Doctor of Philosophy (PhD) in Information Security · Data Protection & Applied Cryptography 2021 – 2025
Kharkiv National University of Internal Affairs (KhNUIA)

Master of Science (MS) in Cybersecurity (With Distinction) 2019 – 2021
Kharkiv National University of Internal Affairs (KhNUIA)

Bachelor of Science (BS) in Cybersecurity 2015 – 2019
Kharkiv National University of Internal Affairs (KhNUIA)

LANGUAGES & SCIENTIFIC CREDENTIALS

Languages: **English** (Professional Working Proficiency / C1 · Full Technical & Executive Communication), **Ukrainian** (Native).
Publications: Author of **20+ peer-reviewed scientific papers** and **1 Ukrainian State Patent** in information security and data compression.